A finite-flux speed limit for Page-like entropy excursions

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Abstract

How rapidly can a complete entropy excursion occur if the outgoing null-energy flux is bounded in magnitude? For two-dimensional conformal vacua, we give the sharp duration bound at fixed entropy height and independent positive and negative flux caps. Its optimizer has two positive-flux intervals enclosing one negative-flux interval, with no distributional energy pulses. The equal-cap problem is an application of the known minimum-time quantum-piston control problem; we make that connection explicit and give a direct proof covering arbitrary additional turnovers. The bound is sharp as an infimum over smooth conformal vacua with the same height and caps. We also derive a necessary lower bound on peak quantum-null-energy-condition nonsaturation from an entropy excursion faster than the conformal-vacuum limit. The result concerns a specified null entropy and affine-time normalization, rather than a general bound on black-hole evaporation.

1 Question and setting

In two-dimensional conformal vacua, the outgoing energy flux and renormalized radiation entropy obey an exact differential relation [1]. This permits entropy histories to constrain energy emission without specifying a bulk evaporation model. Integrated-energy costs of complete Page-like excursions, including their tradeoffs, have recently been optimized [2]; earlier work discussed the energy–entropy relation in evaporation [3]. Here the constraint is instead on the *instantaneous* positive and negative flux. An optimum involving delta-function pulses cannot satisfy such a constraint.

We develop the conformal-field-theory application, state the unequal-cap extension, and derive a nonsaturation diagnostic. The equal-cap control problem was solved by Stefanatos for a quantum piston [4]. Hyperbolic and trigonometric matching of square energy pulses also occurs in quantum-interest analysis [5]. We do not claim a new optimal-control principle or a new derivation of the conformal anomaly.

Set $\hbar = 1$ and let $c > 0$ denote the central charge. For an asymptotically affine ray-tracing map $p(u)$, with $p' > 0$ and $p' = 1$ initially, use

$$S(u) = -\frac{c}{12}\log p'(u), \qquad F(u) = -\frac{c}{24\pi}\{p,u\} = \frac{1}{2\pi}\left(S'' + \frac{6}{c}S'^2\right). \tag{1}$$

Here u is an affinely normalized outgoing null coordinate, primes denote u derivatives, and S is vacuum-subtracted entropy, not an absolute entropy. Defining

$$K = \frac{c}{12\pi}, \quad w = e^{6S/c}, \quad R = e^{6S_{\max}/c} > 1, \qquad F = K\frac{w''}{w}, \quad p' = w^{-2}, \tag{2}$$

consider a complete excursion with $w = 1$ outside $[0,T]$, $1 \le w \le R$, and $\max w = R$. Nonnegative entropy relative to the endpoint value is an explicit assumption. Allow $w \in W^{2,\infty}(\mathbb{R})$, so w, w' are continuous and the flux is essentially bounded. Impose

$$-\mathcal{F}_- \le F \le \mathcal{F}_+, \qquad a = \sqrt{\mathcal{F}_+/K} > 0, \quad b = \sqrt{\mathcal{F}_-/K} > 0. \tag{3}$$

These are externally specified caps, not pointwise energy conditions of QFT.

2 Exact duration bound

Proposition 1. *Every excursion satisfying the above assumptions obeys*

$$\boxed{T \geq \mathcal{T}(a,b;R) = 2\left[\frac{\operatorname{arcosh} y}{a} + \frac{\arccos(y/R)}{b}\right], \qquad y^2 = \frac{b^2R^2 + a^2}{a^2 + b^2}.} \tag{4}$$

The bound is attained in the stated weak class and is the infimum over smooth conformal-vacuum excursions with the same height and caps. Extra turnovers cannot lower it.

Proof. Equation (3) is equivalent to $-b^2 w \leq w'' \leq a^2 w$. On a connected component (α, β) of $\{w' > 0\}$, the endpoint velocities vanish. Put $m = w(\alpha) \geq 1$ and $M = w(\beta) \leq R$. Integrating the two acceleration inequalities, respectively from α and backwards from β, gives

$$w'^2 \leq a^2(w^2 - m^2), \qquad w'^2 \leq b^2(M^2 - w^2). \tag{5}$$

Time reversal gives the same estimates on decreasing components. Thus, everywhere by continuity,

$$|w'| \leq V(w), \qquad V(x) = \min\{a\sqrt{x^2-1},\ b\sqrt{R^2-x^2}\}. \tag{6}$$

Set $H(x) = \int_1^x ds/V(s)$. Its endpoint singularities are integrable. The composition $H(w(u))$ is 1-Lipschitz: this follows first with V replaced by $\max(V, \epsilon)$, using the chain rule, and then by taking $\epsilon \downarrow 0$. If $w(u_*) = R$, then $H(R) \leq u_*$ and $H(R) \leq T - u_*$. Consequently $T \geq 2H(R)$, regardless of the number of stationary points. The branches of (6) cross at y, and

$$H(R) = \int_1^y \frac{dx}{a\sqrt{x^2-1}} + \int_y^R \frac{dx}{b\sqrt{R^2-x^2}} = \frac{\operatorname{arcosh} y}{a} + \frac{\arccos(y/R)}{b}. \tag{7}$$

This proves (4). □

For attainment, write $t_+ = \operatorname{arcosh}(y)/a$, $t_- = \arccos(y/R)/b$, $\tau = t_+ + t_-$, and $T_* = 2\tau$. The complete optimizer is

$$w_*(u) = \begin{cases} 1, & u \notin [0, T_*], \\ \cosh(au), & 0 \leq u \leq t_+, \\ R\cos[b(u-\tau)], & t_+ \leq u \leq T_* - t_+, \\ \cosh[a(T_* - u)], & T_* - t_+ \leq u \leq T_*. \end{cases} \tag{8}$$

The identity $a\sqrt{y^2-1} = b\sqrt{R^2-y^2}$ ensures continuity of w_*'. Its flux is $+\mathcal{F}_+$ on both outer intervals and $-\mathcal{F}_-$ on the central interval of length $2t_-$. Jumps of w_*'' produce finite flux steps, not impulses. Equality in the Lipschitz argument forces $H(w(u)) = \min(u, T_* - u)$; hence the optimizer is unique on its support. The net emitted energy satisfies

$$E_{\text{net}} = 2\mathcal{F}_+ t_+ - 2\mathcal{F}_- t_- = K \int_0^{T_*} (w_*'/w_*)^2 \, du > 0. \tag{9}$$

Smooth sharpness needs care because smoothing lowers the peak. Let $z_\delta = \rho_\delta * w_*$, with a nonnegative smooth mollifier supported in $[-\delta, \delta]$, and set

$$\lambda_\delta = \frac{R-1}{\max z_\delta - 1}, \qquad w_\delta(u) = 1 + \lambda_\delta\left[z_\delta\left(u/\sqrt{\lambda_\delta} - \delta\right) - 1\right]. \tag{10}$$

Convolution preserves $-b^2 z_\delta \leq z_\delta'' \leq a^2 z_\delta$. Here $\lambda_\delta \geq 1$, $w_\delta \geq z_\delta$ at corresponding arguments, and $w_\delta'' = z_\delta''$. Both original caps therefore hold exactly, while $\max w_\delta = R$. Its duration is $\sqrt{\lambda_\delta}(T_* + 2\delta) \to T_*$. Each $p_\delta' = w_\delta^{-2}$ defines a smooth, timelike moving mirror, asymptotically at rest. The final affine intercept of p_δ may differ from the initial one. No globally C^2 excursion attains T_* because (8) has jumps of w''; any duration strictly above T_* is achievable by smoothing and dilation.

3 Peak flux and the size of an excursion

For equal caps $\mathcal{F}_+ = \mathcal{F}_- = \mathcal{F}$, put $x = 6S_{\max}/c$. Equation (4) becomes the sharp peak-flux inequality

$$\boxed{\mathcal{F} \geq \frac{c}{3\pi T^2} h(x)^2, \qquad h(x) = \frac{1}{2}\operatorname{arcosh}(e^{2x}) + \arctan\sqrt{\tanh x}.} \tag{11}$$

The two terms are the acceleration and braking times of the quantum-piston solution [4], after translating piston expansion into exponentiated conformal entropy. In particular,

$$h(x) = 2\sqrt{x} + \tfrac{1}{15}x^{5/2} + O(x^{9/2}), \qquad x \downarrow 0, \tag{12}$$

$$h(x) = x + \tfrac{1}{2}\log 2 + \tfrac{\pi}{4} + O(e^{-2x}), \qquad x \to \infty. \tag{13}$$

Thus small excursions require $\mathcal{F}_{\min} \sim 8S_{\max}/(\pi T^2)$, whereas large excursions have $\mathcal{F}_{\min} \sim 12S_{\max}^2/(\pi c T^2)$. The change from linear to quadratic dependence distinguishes a finite peak constraint from merely bounding total emitted energy.

As a nonoptimal comparison, take $S = (c/6)x\sin^2(\pi u/T)$ on $[0, T]$, with constant continuation. Its flux is

$$\frac{F}{K} = \frac{\pi^2}{T^2}\left[2x\cos(2\pi u/T) + x^2\sin^2(2\pi u/T)\right]. \tag{14}$$

Its largest absolute flux, divided by the optimum (11), is

$$\frac{\mathcal{F}_{\sin^2}}{\mathcal{F}_{\min}} = \frac{\pi^2}{4h(x)^2}\begin{cases} 2x, & 0 < x \leq 1, \\ x^2 + 1, & x \geq 1. \end{cases} \tag{15}$$

The ratio tends to $\pi^2/8$ for small excursions and $\pi^2/4$ for large ones. This comparison concerns peak flux; it is different from the integrated-energy sinusoidal comparison in Ref. [2].

Unequal caps expose two complementary obstructions:

$$\lim_{b\to\infty}\mathcal{T} = \frac{2\operatorname{arcosh} R}{a}, \qquad \lim_{a\to\infty}\mathcal{T} = \frac{2\arccos(1/R)}{b}. \tag{16}$$

At fixed positive cap, vanishing negative cap sends the duration to infinity. At fixed negative cap, allowing arbitrarily intense positive pulses leaves a nonzero braking time. The one-sided limits admit impulses of the unconstrained sign and should be interpreted as limits, not finite-cap optimizers.

4 A diagnostic for QNEC nonsaturation

For a state and null deformation obeying the strengthened two-dimensional quantum null energy condition (QNEC), write

$$F = F_S + \Delta_Q, \qquad F_S = \frac{1}{2\pi}\left(S'' + \frac{6}{c}S'^2\right), \qquad 0 \leq \Delta_Q \leq D. \tag{17}$$

Here the condition must hold for the same entropy S used above, for example in a half-line limit where the reference entropy has vanishing deformation derivatives. Generic finite-interval vacuum subtraction cannot be inserted into this nonlinear condition without justification. Saturation and nonsaturation of this condition distinguish physically different classes of two-dimensional states [6]. With the same entropy boundary conditions and measured flux caps, (17) implies

$$-(\mathcal{F}_- + D) \leq F_S \leq \mathcal{F}_+. \tag{18}$$

The mathematical proposition therefore gives the necessary condition

$$T \geq \mathcal{T}\left(\sqrt{\mathcal{F}_+/K}, \sqrt{(\mathcal{F}_- + D)/K}; R\right). \tag{19}$$

For fixed a, R, $\mathcal{T}(a, b; R)$ is strictly decreasing in b: in (6) the braking envelope increases on a nonempty interval. If $2\operatorname{arcosh}(R)/a < T < \mathcal{T}(a, \sqrt{\mathcal{F}_-/K}; R)$, there is a unique $b_{\text{crit}} > \sqrt{\mathcal{F}_-/K}$ solving $\mathcal{T}(a, b_{\text{crit}}; R) = T$. Consequently

$$\operatorname{ess\,sup}\Delta_Q \;\geq\; K b_{\text{crit}}^2 - \mathcal{F}_- > 0. \tag{20}$$

An excursion at or below the positive-cap floor $2\,\mathrm{arcosh}(R)/a$ is incompatible with a regular profile under these assumptions: equality would require an instantaneous reversal of nonzero w' at the maximum. For durations at or above the zero-gap bound this diagnostic supplies no positive lower bound. Equation (20) is a necessary witness, not a claim that every pair (S, F) obeying QNEC is realized by a quantum state, nor a claim of sharpness in the nonsaturating sector. For a smooth profile a positive threshold in (20) is a strict inequality, since equality would force (8).

The diagnostic becomes strong near the positive-cap floor $T_{\mathrm{f}} = 2\,\mathrm{arcosh}(R)/a$. Expanding (4) at large b gives

$$\mathcal{T} = T_{\mathrm{f}} + \frac{a\sqrt{1-R^{-2}}}{b^2} + O(b^{-4}), \qquad D_{\mathrm{crit}} \sim \frac{Ka\sqrt{1-R^{-2}}}{T - T_{\mathrm{f}}}. \tag{21}$$

For example, $R = e$ and $\mathcal{F}_+ = \mathcal{F}_- = K$ give $T_{\mathrm{f}} = 3.314909$ and a zero-gap duration 4.123581. An excursion of duration 3.5 requires $D \geq 3.853457K$. This is a conditional diagnostic, not an example of a constructed nonsaturating state.

5 Scope and interpretation

The result supplies a benchmark for conformal-vacuum Page-like pulses with finite stress: it fixes the optimal duration, the switch locations, and the minimum peak magnitude. Its possible use beyond that sector is the conditional QNEC-gap diagnostic. The formulas refer to the same affine null coordinate throughout. Under $\widetilde{u} = \lambda u$, $\widetilde{T} = \lambda T$ and $\widetilde{\mathcal{F}}_\pm = \mathcal{F}_\pm/\lambda^2$, so the bound is covariant under this rescaling.

The flux caps are assumptions on expectation values and do not follow from quantum energy inequalities, which constrain smeared energy [7]. Neither entropy undershoots below the endpoint value nor thermal backgrounds are covered. The constructed mirror is a prescribed boundary, not a solution of gravitational backreaction. In particular, an island formula for black-hole entropy cannot simply be substituted into (1). These restrictions leave a concrete CFT optimization and diagnostic; they do not establish a universal evaporation-time bound.

References

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